IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

[Your Name]
October 25, 2026



Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

Methodologies

- Collected launch data from the SpaceX REST API and Falcon 9 history scraped from Wikipedia
- Wrangled the data into a binary Class label (1 = first stage landed, 0 = failed)
- EDA with Matplotlib/Seaborn and SQL; interactive analytics with Folium and Plotly Dash
- Built, tuned (GridSearchCV) and compared Logistic Regression, SVM, Decision Tree and KNN

Results

- Landing success grew from 0% in 2010-2013 to about 80%+ after 2016 as boosters were reused
- KSC LC-39A has the best success ratio (about 77%); orbits ES-L1, GEO, HEO and SSO show 100%
- All four models reach 83% test accuracy; the Decision Tree has the best cross-validation score

~67%
of Falcon 9 landings succeeded (60 of 90 launches)

83%
best model test accuracy

4
launch sites analysed

4
ML models compared

Business value
Predicting first stage reuse lets a competitor estimate launch cost (about $62M vs. $165M+ when the booster is not reused) and bid against SpaceX.

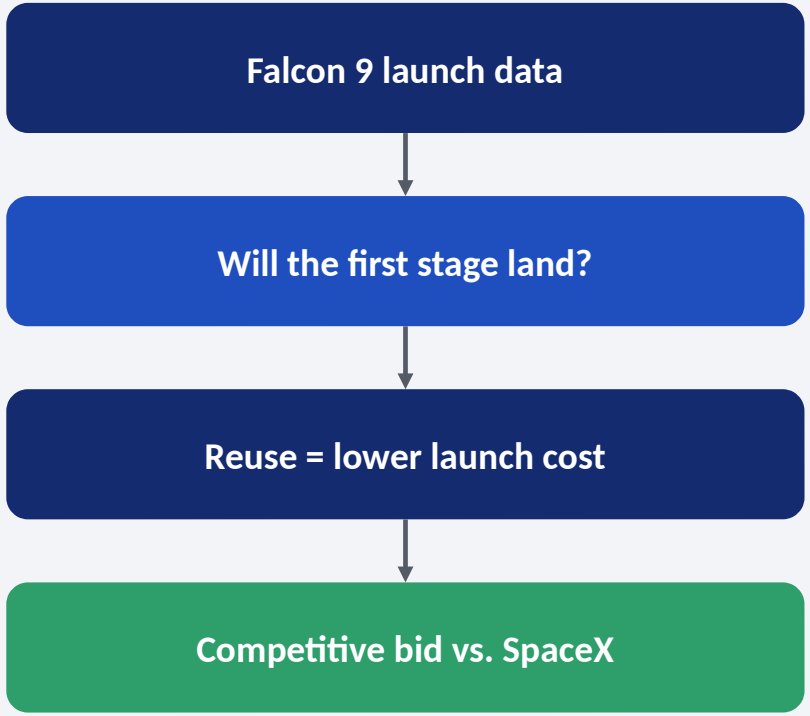
Introduction

Project background and context

- SpaceX advertises Falcon 9 launches at about $62M, versus $165M+ for competitors; the saving comes from reusing the first stage
- If we can predict whether the first stage will land, we can estimate the real cost of a launch
- Company "Space Y" wants to compete with SpaceX and needs that insight

Problems we want to answer

- Which factors (launch site, payload, orbit, booster version, year) drive landing success?
- How has the success rate changed over time?
- Which classification model best predicts whether the first stage lands?



Section 1

Methodology



Methodology

Data collection

- SpaceX REST API (past launches) and web scraping of the Wikipedia Falcon 9 launch table

Data wrangling

- Cleaned missing values, filtered Falcon 9, created the landing Class label

Exploratory data analysis (EDA) using visualization and SQL

Interactive visual analytics using Folium and Plotly Dash

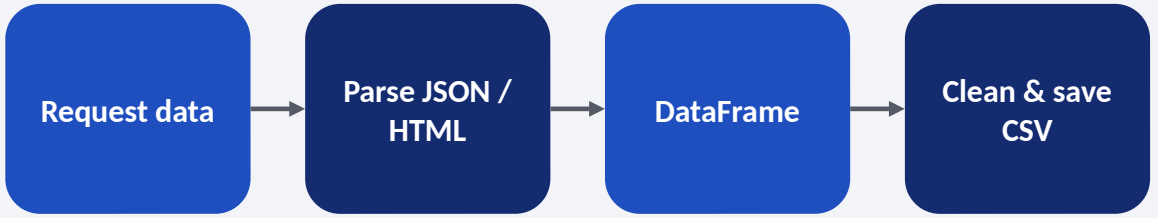
Predictive analysis using classification models

- Standardise, split 80/20, tune with GridSearchCV (cv=10), evaluate on the test set

Data Collection

Two complementary sources were combined

- SpaceX REST API: flight number, date, booster version, payload mass, orbit, launch site, outcome, reuse and landing-leg details
- Wikipedia scraping (BeautifulSoup): Falcon 9 launch records used to validate and enrich the data
- Both were converted into Pandas DataFrames for wrangling



Output: dataset_part_1.csv (API) and a scraped launch table, ready for wrangling.

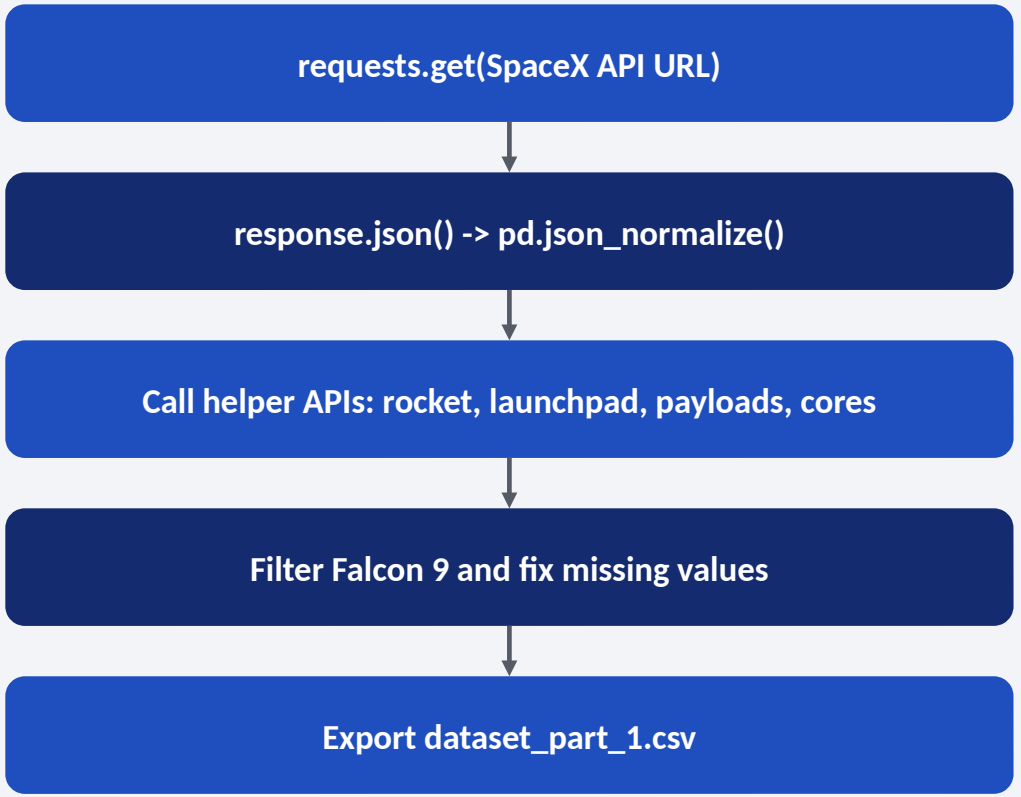
Data Collection – SpaceX API

GET /v4/launches/past returns raw JSON

- Helper functions look up booster, launch pad, payload and core IDs
- Normalised JSON into a DataFrame and kept only Falcon 9 launches
- Replaced missing payload mass with the mean

GitHub notebook:

- [Paste your GitHub URL here]



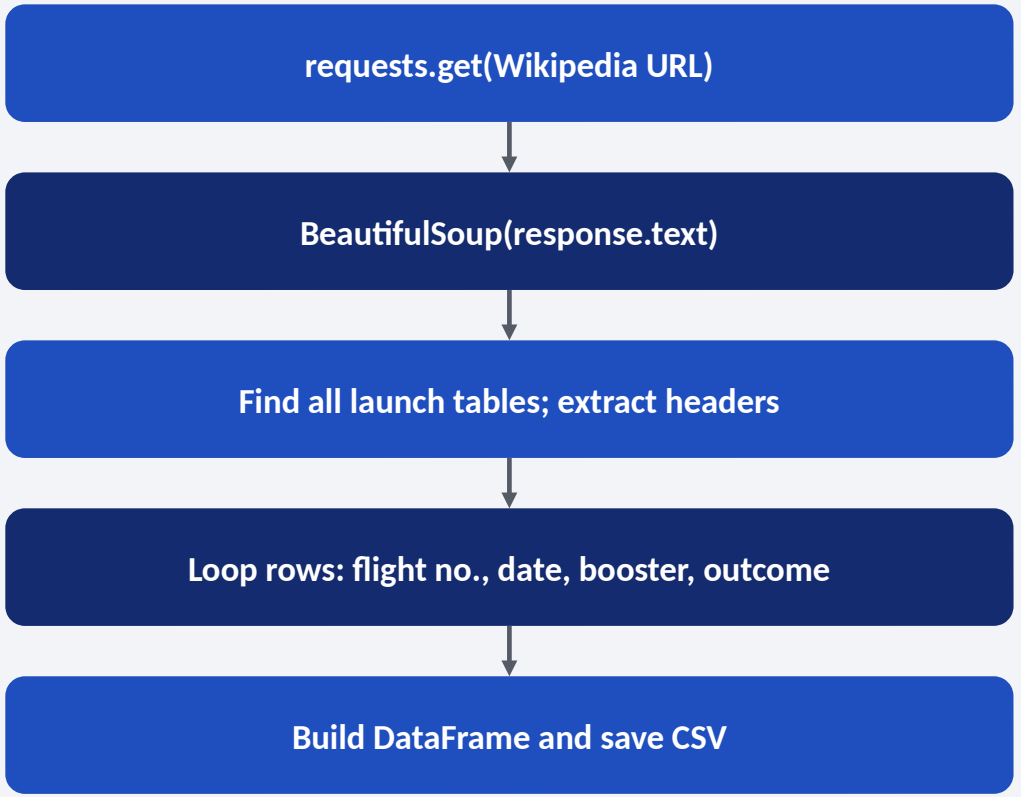
Data Collection - Scraping

Scraped the "List of Falcon 9 and Falcon Heavy launches" Wikipedia page

- BeautifulSoup parses the HTML tables
- Extracted column names, then each launch row
- Combined into one DataFrame and saved to CSV

GitHub notebook:

- [Paste your GitHub URL here]



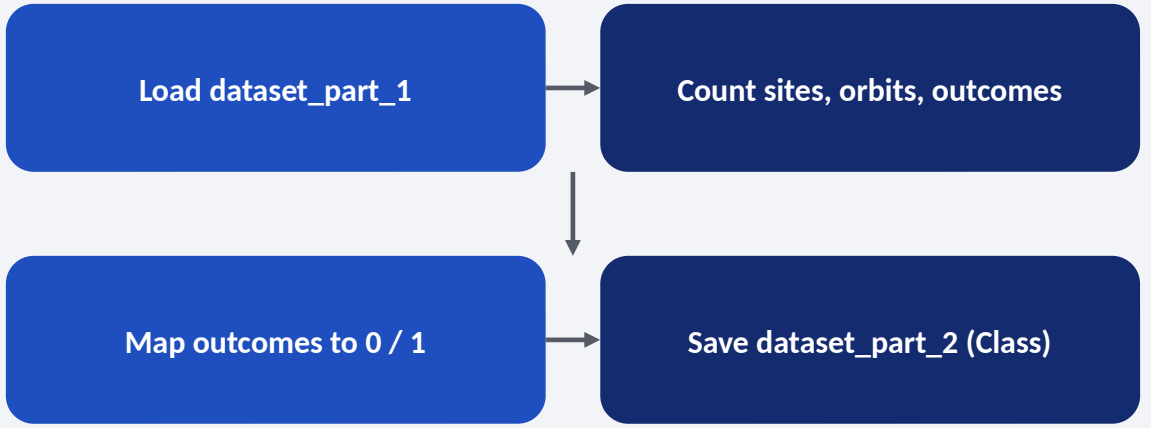
Data Wrangling

Goal: turn the landing outcome into a label

- Counted launches per site, orbit and mission outcome
- Outcomes True ASDS, True RTLS, True Ocean = success (1)
- False ASDS, False RTLS, False Ocean, None = failure (0)
- Result: Class column; overall success rate about 66.7% (60 of 90)

GitHub notebook:

- [Paste your GitHub URL here]



Class = 1 for a landed first stage, 0 otherwise; this is the target for classification.

EDA with Data Visualization

Charts plotted and why

- Scatter: Flight Number vs. Launch Site and Payload vs. Launch Site, to see site experience and payload limits
- Bar: success rate per orbit, to compare orbit difficulty
- Scatter: Flight Number and Payload vs. Orbit, to see how orbit mix changed
- Line: yearly average success rate, to show learning over time
- Feature engineering: one-hot encoded categorical features

GitHub notebook:

- [Paste your GitHub URL here]

EDA with SQL

SQL queries run (SQLite via ipython-sql) on the SPACEXTBL table

- Unique launch sites (SELECT DISTINCT)
- Five records where site starts with CCA (LIKE)
- Total and average payload (SUM, AVG) for NASA CRS and F9 v1.1
- First successful ground pad landing (MIN date)
- Drone ship successes with payload 4000-6000 kg
- Success vs. failure counts (GROUP BY)
- Max payload boosters (subquery) and 2015 drone ship failures
- Ranked landing outcomes 2010-06-04 to 2017-03-20 (ORDER BY ... DESC)

GitHub notebook:

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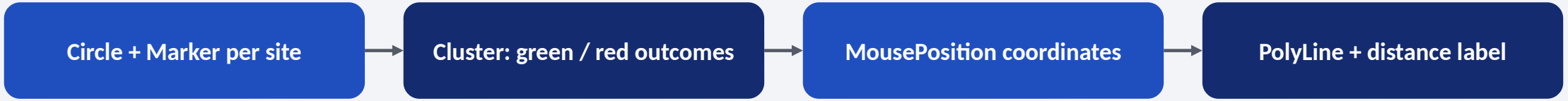
Build an Interactive Map with Folium

Map objects created

- Circle + Marker at each launch site (NASA JSC as reference): shows exactly where the sites are
- MarkerCluster of launches, green = success, red = failure: shows which sites succeed most
- MousePosition: reads coordinates of coastline, railway and highway
- PolyLine + distance label from a site to each proximity: quantifies safety and logistics distance

GitHub notebook:

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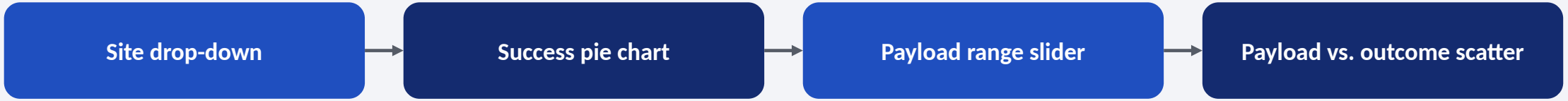
Build a Dashboard with Plotly Dash

Dashboard components

- Drop-down to choose a launch site (or all sites): lets users focus on one site
- Pie chart of successful launches: compares sites, or success vs. failure for one site
- Range slider for payload mass (0-10,000 kg): filters launches by payload
- Scatter plot of payload vs. outcome coloured by booster version: shows which payloads and boosters succeed
- Callbacks link the drop-down and slider to both charts

GitHub notebook / app:

- [Paste your GitHub URL here]



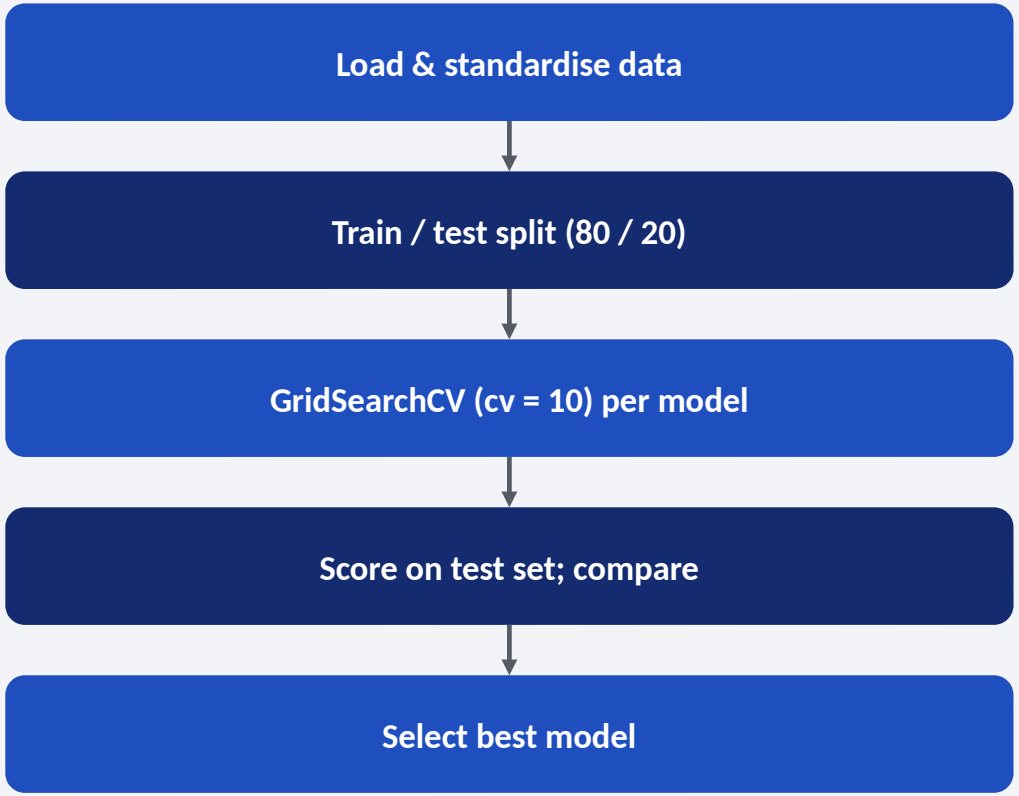
Predictive Analysis (Classification)

Model development

- Standardised features (StandardScaler) and split 80/20 (random_state=2)
- GridSearchCV with cv=10 for Logistic Regression, SVM, Decision Tree and KNN
- Evaluated test accuracy and confusion matrices
- Best model: Decision Tree (highest CV accuracy); all models tie at 83% on the test set

GitHub notebook:

- [Paste your GitHub URL here]



Results

Exploratory data analysis results

- Success improved steadily with experience; KSC LC-39A and orbits such as SSO, GEO and HEO perform best

Interactive analytics demo in screenshots

- Launch sites sit near the coast and far from cities; the dashboard confirms KSC LC-39A leads

Predictive analysis results

- Classification models reach about 83% accuracy; errors are mostly false positives

0% to 80%+
landing success, 2010 to recent

77%
KSC LC-39A success ratio

83%
best model test accuracy

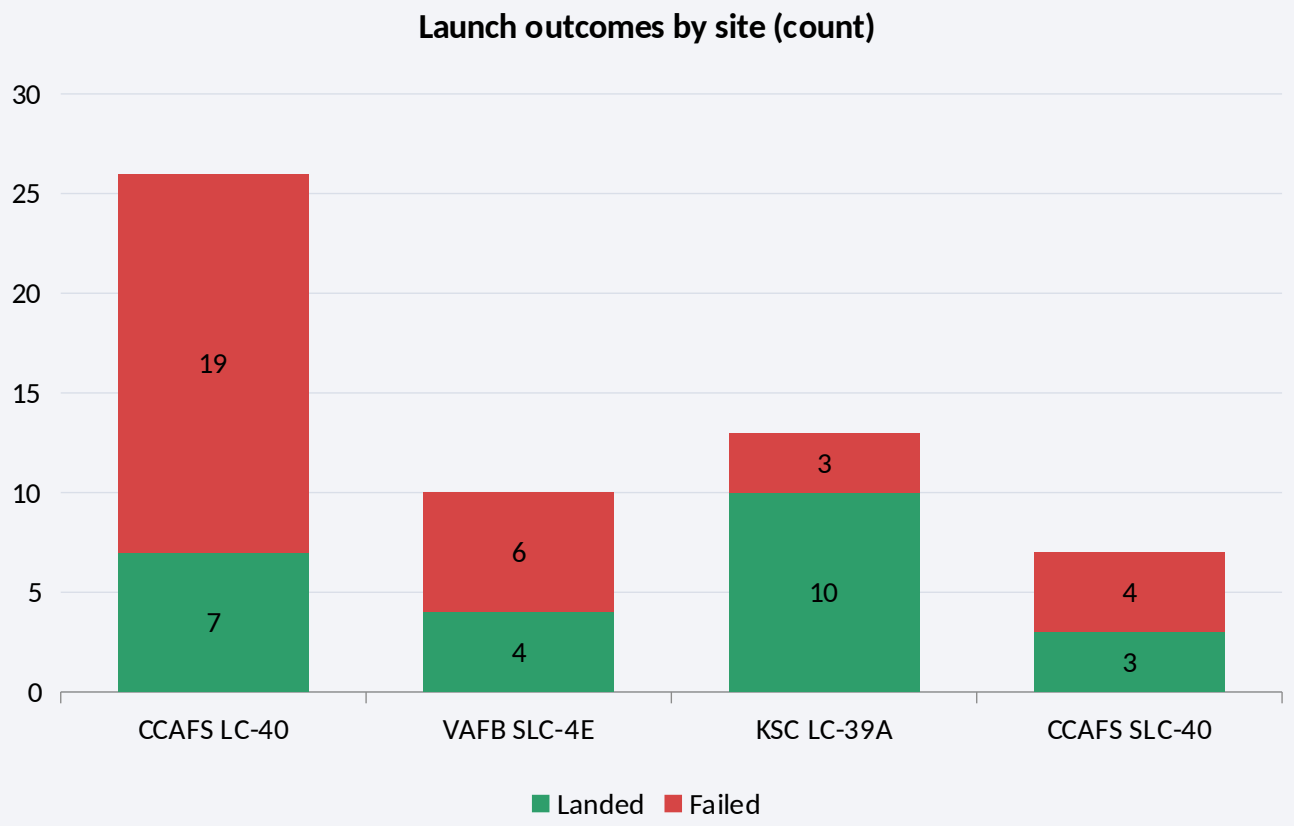
Section 2

Insights drawn from EDA

Flight Number vs. Launch Site

Scatter of Flight Number vs. Launch Site, coloured by outcome

- Success rate rises with flight number at every site
- Early flights (below ~20) mostly failed; later flights mostly landed
- CCAFS LC-40 has the most launches; KSC LC-39A is used for later flights



Payload vs. Launch Site

Payload mass vs. Launch Site

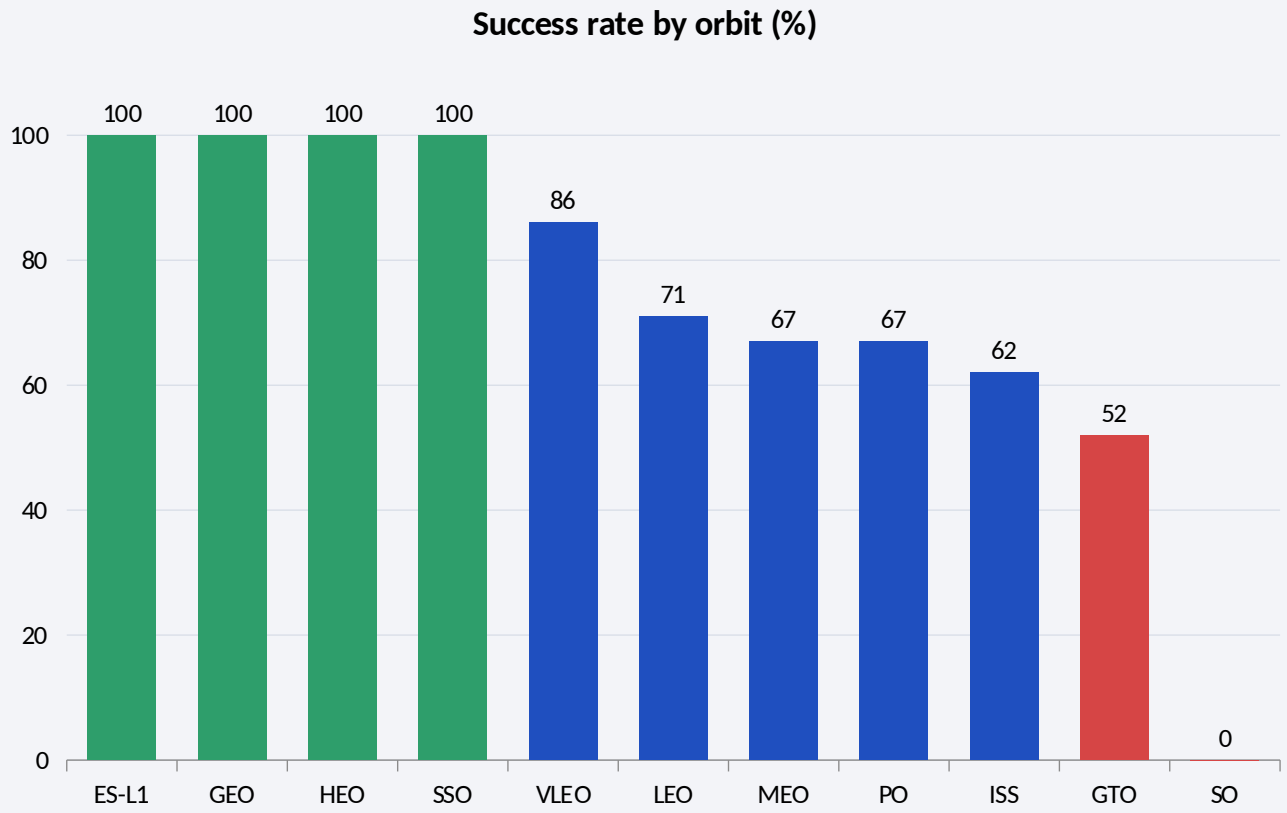
- VAFB SLC-4E has no launches above ~10,000 kg (polar orbit missions)
- Heavy payloads (over 7,000 kg) land more reliably than light ones
- KSC LC-39A and CCAFS carry the heaviest payloads

Paste your Payload vs. Launch Site scatter plot (from the EDA notebook) here

Success Rate vs. Orbit Type

Success rate of each orbit type

- ES-L1, GEO, HEO and SSO: 100% success (few launches each)
- VLEO 86%, LEO 71%, MEO / PO 67%, ISS 62%
- GTO is hardest at about 52%; SO has 0%



Flight Number vs. Orbit Type

Flight Number vs. Orbit Type

- In LEO the success rate relates to the number of flights
- For GTO there is no clear relation between flight number and success
- Early flights were mainly LEO and ISS; GTO and SSO missions came later

Paste your Flight Number vs. Orbit scatter plot (from the EDA notebook) here

Payload vs. Orbit Type

Payload vs. Orbit Type

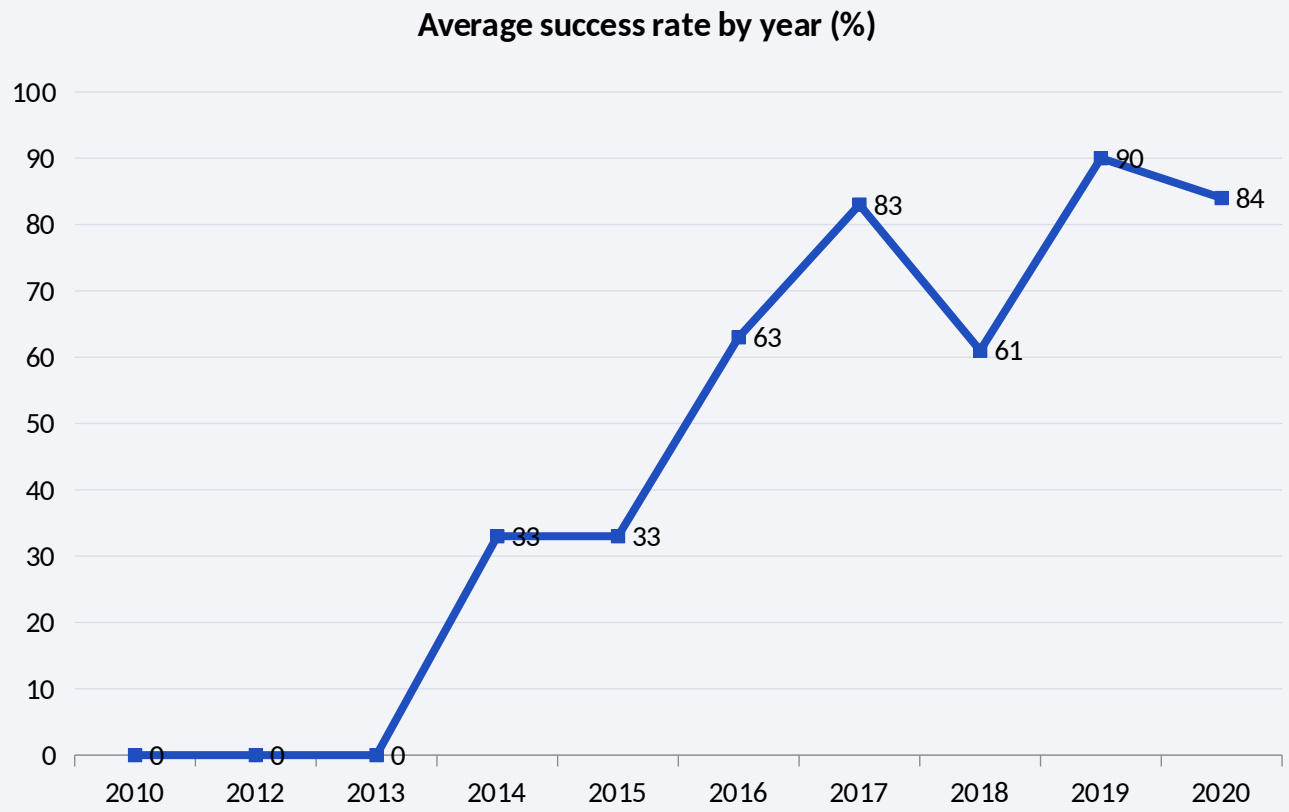
- With heavy payloads, successful landings are more common in PO, LEO and ISS
- GTO shows both outcomes across payload sizes, so payload alone cannot separate them
- Orbit and payload together are useful model features

Paste your Payload vs. Orbit scatter plot (from the EDA notebook) here

Launch Success Yearly Trend

Yearly average success rate

- No successful landings from 2010 to 2013
- Success rose from 2013 and improved sharply after 2016
- About 80%+ in recent years, showing booster recovery maturing



All Launch Site Names

Find the names of the unique launch sites

- DISTINCT removes duplicates, leaving four sites: three in Florida (CCAFS, KSC) and one in California (VAFB).

```
SELECT DISTINCT Launch_Site FROM SPACEXTBL;
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

Find 5 records where launch sites begin with 'CCA'

- LIKE with the wildcard % filters site names starting with CCA; LIMIT 5 returns the first five, all early test flights.

```
SELECT * FROM SPACEXTBL WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

Date	Booster_Version	Launch_Site	Payload	Mass (kg)	Outcome
2010-06-04	F9 v1.0 B0003	CCAFS LC-40	Dragon Qualification Unit	0	Failure (parachute)
2010-12-08	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1	0	Failure (parachute)
2012-05-22	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	No attempt
2012-10-08	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	No attempt
2013-03-01	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	No attempt

Total Payload Mass

Total payload carried by boosters from NASA

- SUM adds the payload mass of all NASA CRS missions: 45,596 kg were carried for NASA.

```
SELECT SUM(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Customer = 'NASA (CRS)';
```

Total payload (kg)
45,596

Average Payload Mass by F9 v1.1

Average payload mass carried by booster version F9 v1.1

- F9 v1.1 boosters carried an average of about 2,928 kg per launch.

```
SELECT AVG(PAYLOAD_MASS__KG_) FROM SPACEXTBL WHERE Booster_Version = 'F9 v1.1';
```

Average payload (kg)
2,928.4

First Successful Ground Landing Date

Date of the first successful landing on a ground pad

- MIN(Date) over successful ground pad landings gives 22 December 2015, a historic first.

```
SELECT MIN(Date) FROM SPACEXTBL WHERE Landing_Outcome = 'Success (ground pad)';
```

First ground pad success
2015-12-22

Successful Drone Ship Landing with Payload between 4000 and 6000

Boosters that landed on a drone ship with payload between 4,000 and 6,000 kg

- Four Full Thrust boosters landed successfully on a drone ship with mid-range payloads.

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL
WHERE Landing_Outcome = 'Success (drone ship)'
AND PAYLOAD_MASS__KG_ > 4000 AND PAYLOAD_MASS__KG_ < 6000;
```

Booster_Version
F9 FT B1022
F9 FT B1026
F9 FT B1021.2
F9 FT B1031.2

Total Number of Successful and Failure Mission Outcomes

Total number of successful and failure mission outcomes

- Almost every mission itself succeeded (100 of 101); only one failed in flight, so mission success differs from landing success.

```
SELECT Mission_Outcome, COUNT(*) FROM SPACEXTBL GROUP BY Mission_Outcome;
```

Mission_Outcome	Count
Success	100
Failure (in flight)	1

Boosters Carried Maximum Payload

Boosters that carried the maximum payload mass

- A subquery finds the heaviest payload (15,600 kg); the boosters that carried it are all Block 5.

```
SELECT DISTINCT Booster_Version FROM SPACEXTBL WHERE PAYLOAD_MASS__KG_ =
(SELECT MAX(PAYLOAD_MASS__KG_) FROM SPACEXTBL);
```

Booster_Version (max payload 15,600 kg)
F9 B5 B1048.4, B1049.4, B1051.3, B1056.4
F9 B5 B1048.5, B1049.5, B1060.2, B1058.3 and others

2015 Launch Records

Failed drone ship landings in 2015 with booster versions and launch site

- There were two failed drone ship attempts in 2015, both on F9 v1.1 boosters from CCAFS LC-40.

```
SELECT substr(Date,6,2) AS Month, Landing_Outcome, Booster_Version, Launch_Site
FROM SPACEXTBL WHERE Landing_Outcome = 'Failure (drone ship)' AND substr(Date,0,5)='2015';
```

Month	Landing_Outcome	Booster_Version	Launch_Site
01	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
04	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

Rank landing outcomes between 2010-06-04 and 2017-03-20

- No attempt dominates the early years; drone ship successes then equal failures, and ground pad landings follow.

```
SELECT Landing_Outcome, COUNT(*) AS cnt FROM SPACEXTBL
WHERE Date BETWEEN '2010-06-04' AND '2017-03-20'
GROUP BY Landing_Outcome ORDER BY cnt DESC;
```

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3



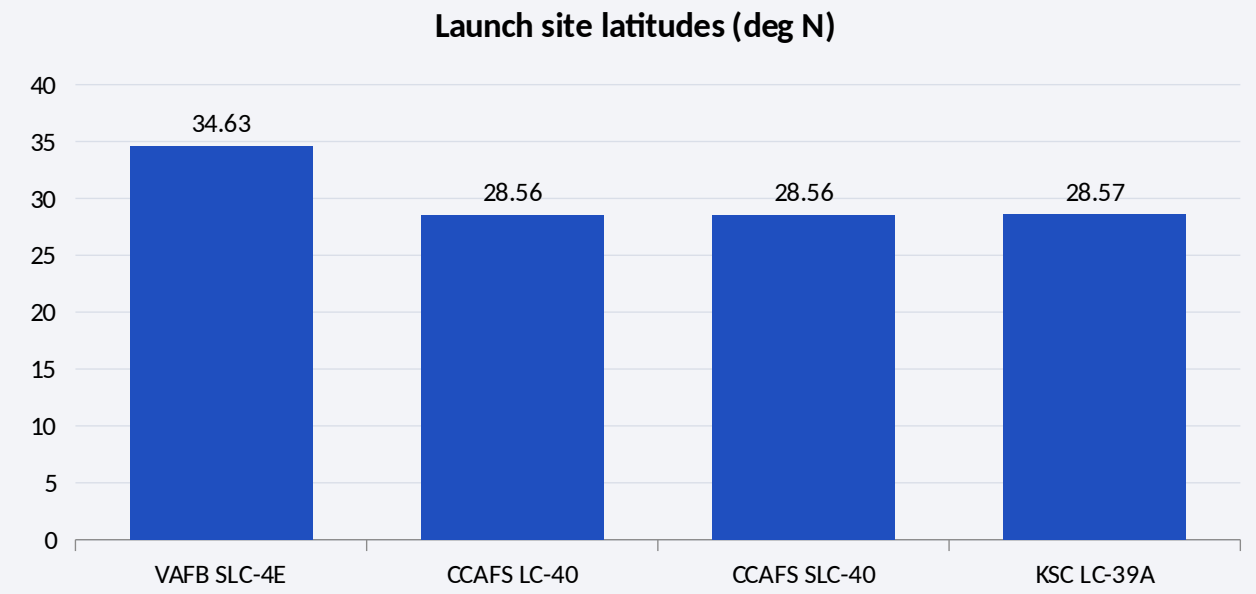
Section 3

Launch Sites Proximities Analysis

All Launch Sites on a Global Map

Each site has a Circle and a Marker label

- All sites are in the US near the equator-facing coast; Earth rotation gives extra speed
- Three sites are close together in Florida; one is on the California coast
- Every site is next to the ocean



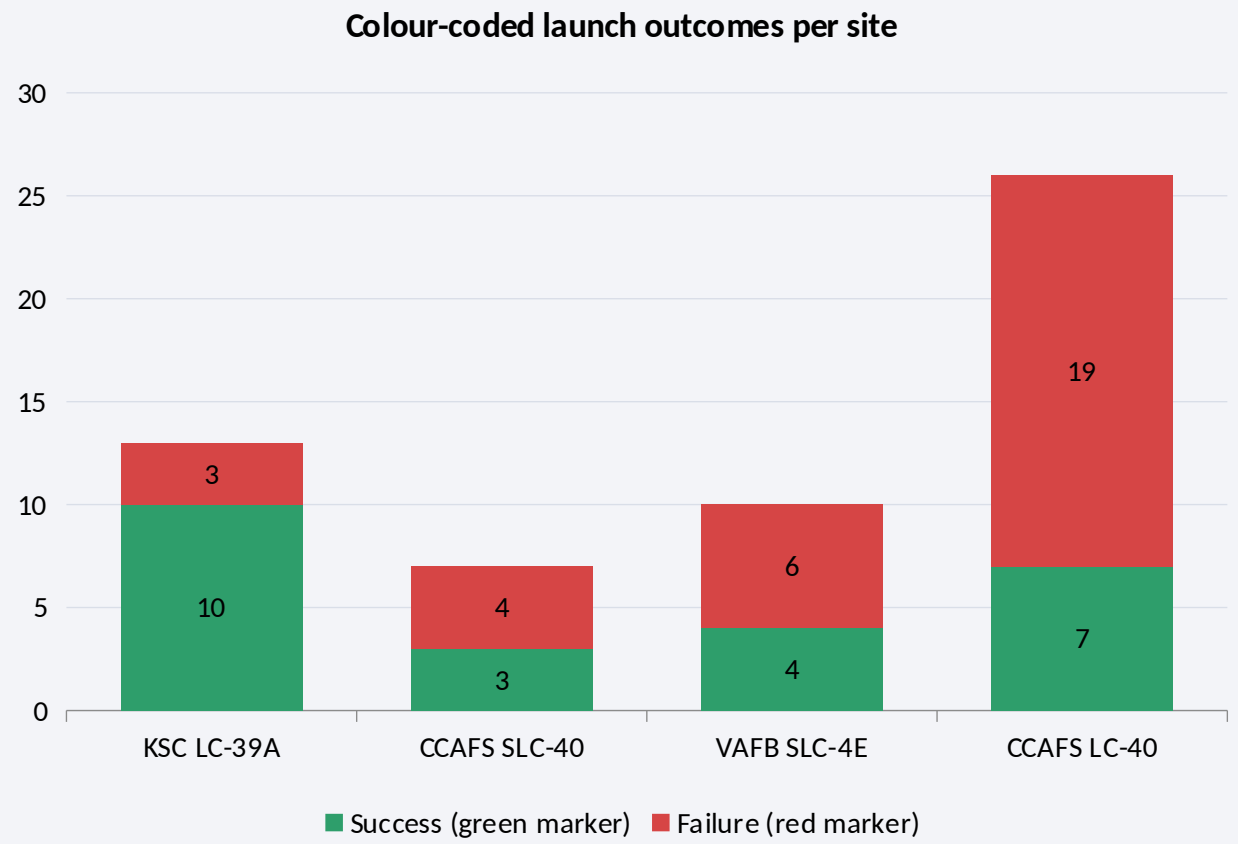
Florida cluster
CCAFS LC-40, CCAFS SLC-40, KSC LC-39A
near -80.6 deg W

California site
VAFB SLC-4E at -120.6 deg W

Colour-Labeled Launch Outcomes

Green markers = success, red = failure in each cluster

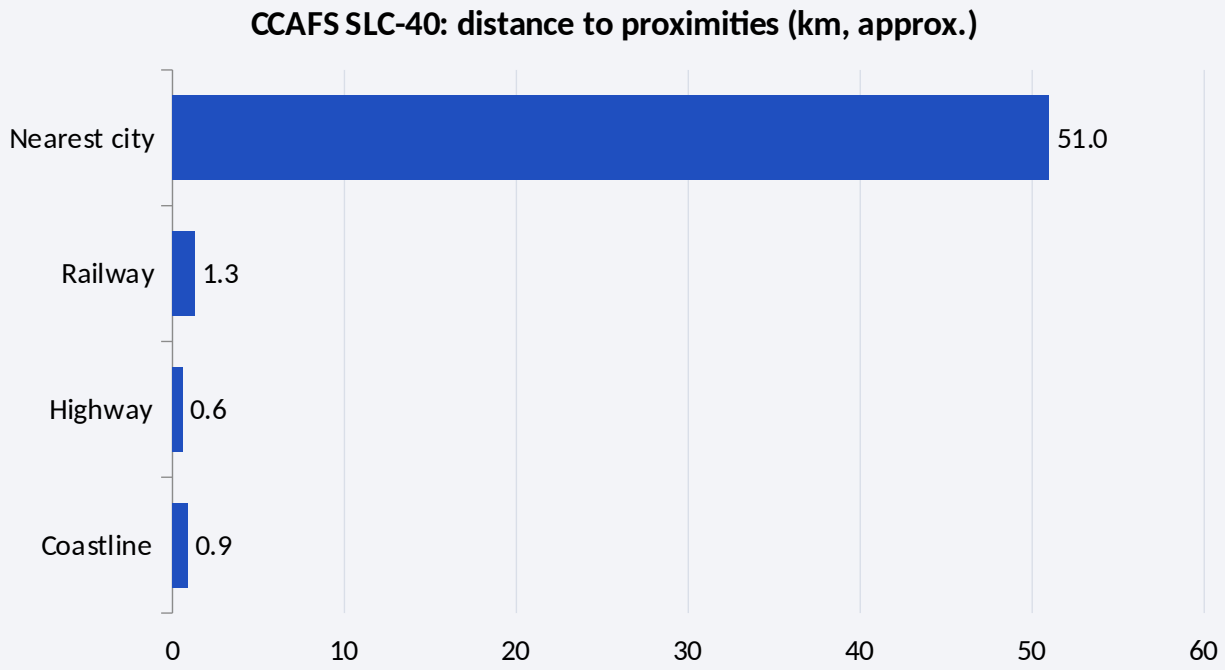
- KSC LC-39A is mostly green: highest success rate
- CCAFS LC-40 is mostly red: many early failed flights
- Colour coding makes site performance obvious at a glance



Launch Site Proximities and Distances

PolyLines with distance labels from the selected site

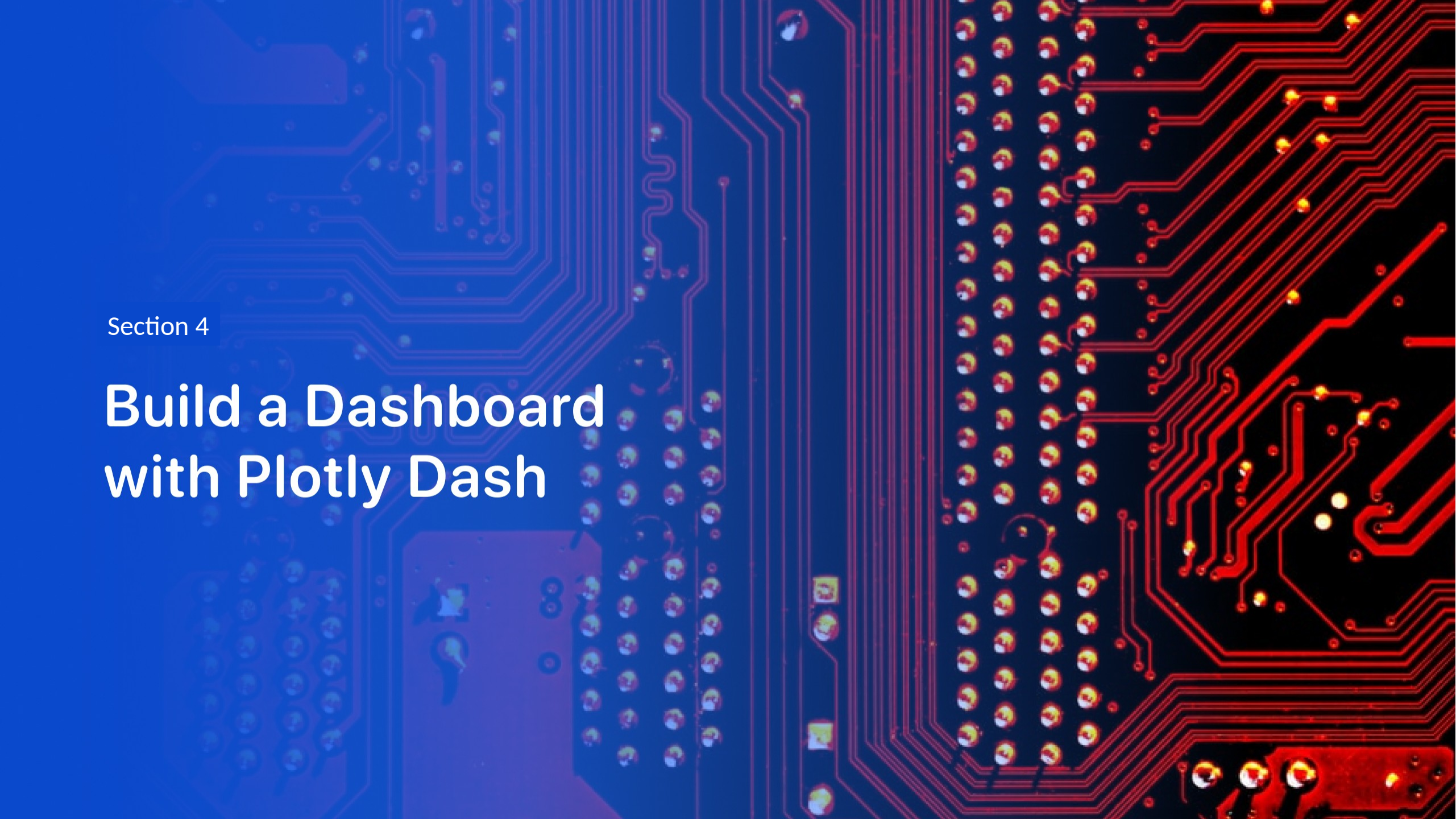
- Coastline: under 1 km, so rockets fly over water
- Highway and railway are close, easing transport of equipment
- Nearest city is about 50 km away, a safety buffer



Close to coast, highway and railway for logistics and safe ocean trajectories, yet far from cities.

Section 4

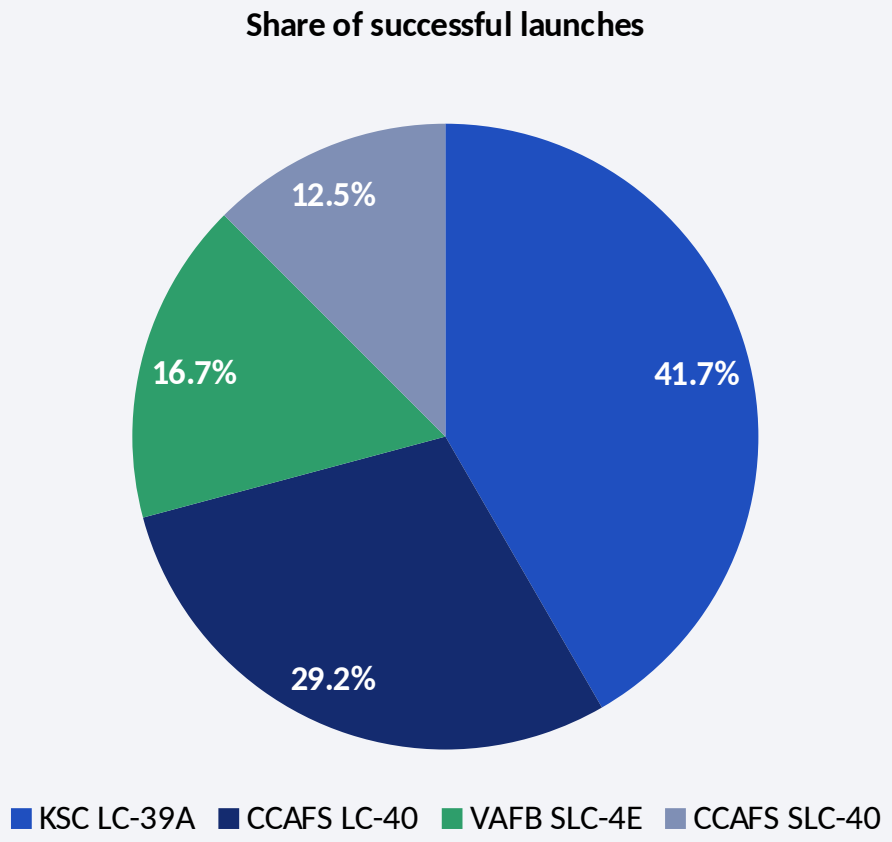
Build a Dashboard with Plotly Dash



Launch Success Count for All Sites

Pie chart of successful launches per site

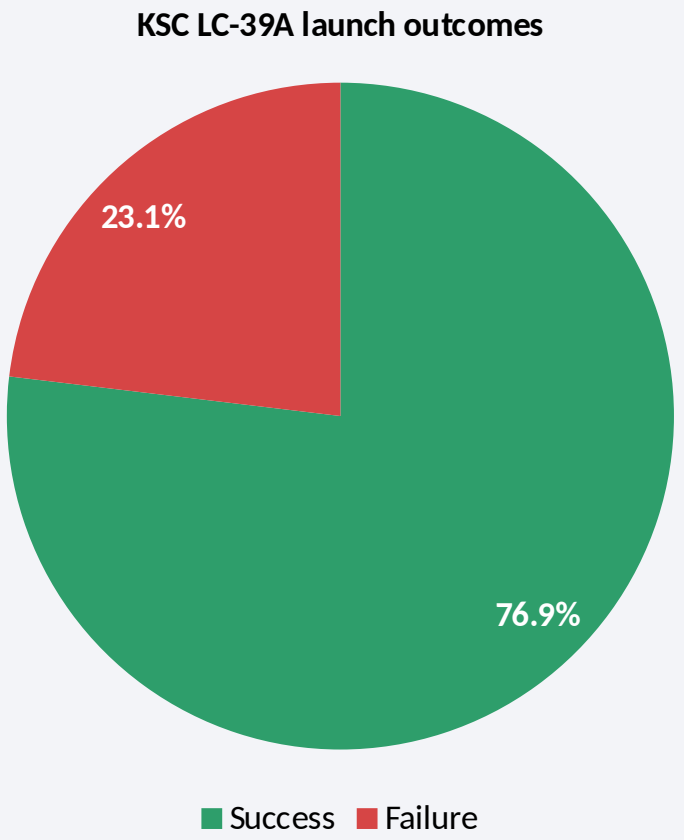
- KSC LC-39A contributes 41.7% of all successes
- CCAFS LC-40 follows with 29.2%
- VAFB SLC-4E 16.7% and CCAFS SLC-40 12.5%



Site with the Highest Success Ratio: KSC LC-39A

Pie for KSC LC-39A only

- 76.9% of launches landed (10 of 13)
- 23.1% failed (3 of 13)
- Best success ratio of all four sites



Payload vs. Launch Outcome

Scatter plot with payload range slider

- Payloads of about 2,000-5,500 kg have the highest success rate
- The FT booster version has the most successes
- Heaviest payloads (over 6,000 kg) have a lower rate

Paste your Payload vs. Launch Outcome dashboard screenshot here

Section 5

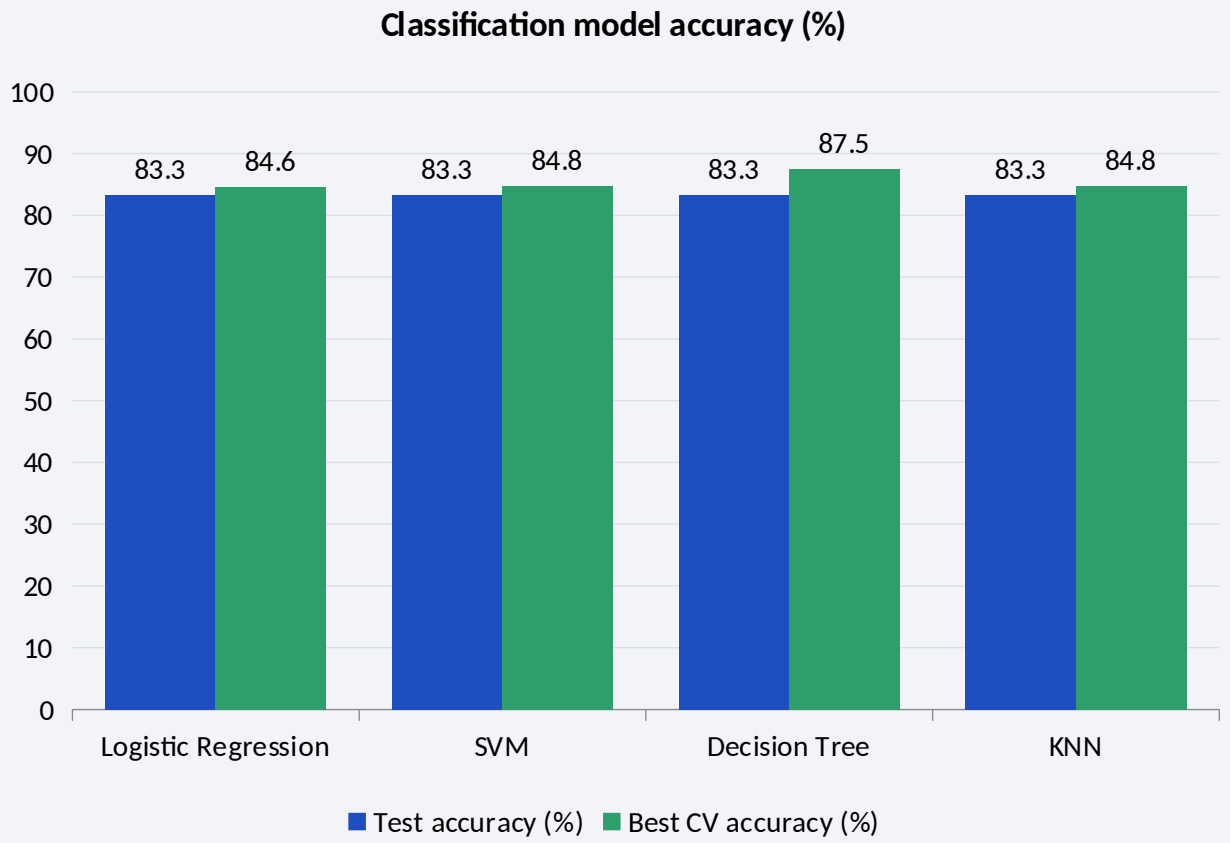
Predictive Analysis (Classification)



Classification Accuracy

Test-set accuracy of all four models

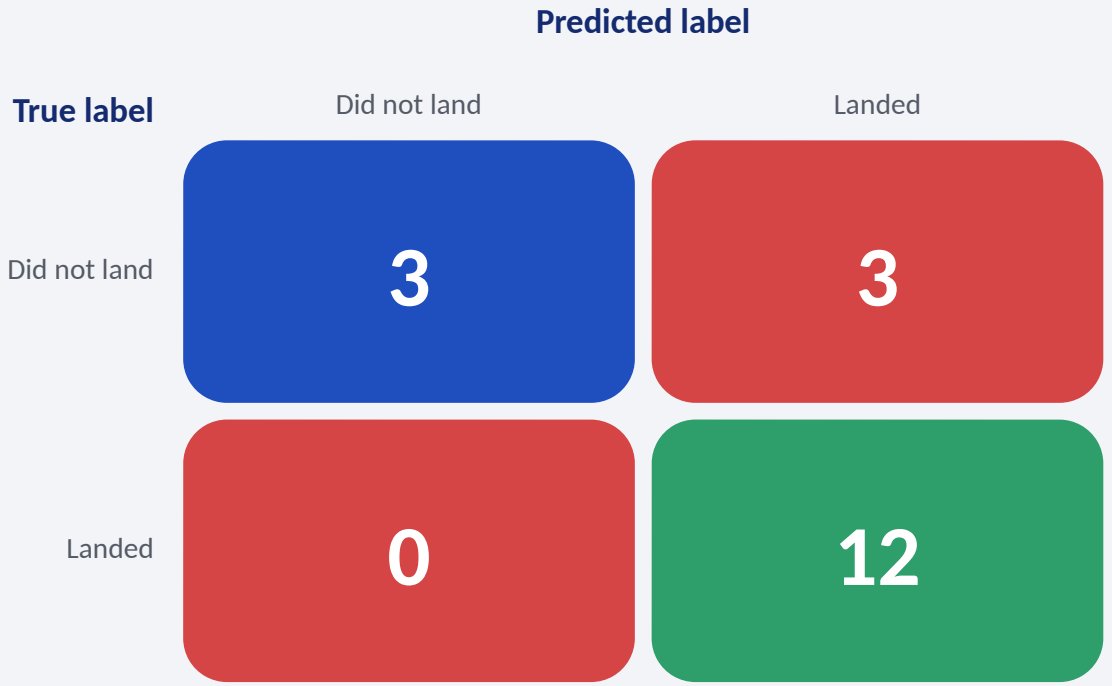
- Logistic Regression, SVM, Decision Tree and KNN each reach 83.3% on 18 test samples
- Decision Tree has the best cross-validation accuracy (about 87-89%)
- Best performing model: Decision Tree



Confusion Matrix

Confusion matrix of the best model (18 test launches)

- 12 landings and 3 non-landings predicted correctly
- 0 missed landings (no false negatives)
- 3 false positives: predicted to land but did not
- Main weakness: over-predicting success



Conclusions

Landing success rose from 0% (2010-13) to about 80% + in recent years

KSC LC-39A has the best site record (about 77%); VAFB and CCAFS lag

Orbits ES-L1, GEO, HEO and SSO show 100% success; GTO is hardest (about 52%)

Payloads of 2,000-5,500 kg and Full Thrust / Block 5 boosters succeed most

All four classifiers reach about 83% accuracy; the Decision Tree is best

Next steps: more launch data, extra features (weather, reuse count) and more tuning

0% to 80%+

success rate, 2010 to recent

83%

best model accuracy

KSC LC-39A

best launch site (77%)

Tree

best classifier: Decision Tree

Insight: experience matters most. Each year of reuse learning raised landing success, so a bidder should price launches by launch-era maturity rather than payload alone.

Appendix

Key code and queries from the project

- Standardise and split: X = preprocessing.StandardScaler().fit(X).transform(X); train_test_split(X, Y, test_size=0.2, random_state=2)
- Tune: GridSearchCV(estimator, parameters, cv=10).fit(X_train, Y_train)
- Evaluate: model.score(X_test, Y_test); confusion_matrix(Y_test, model.predict(X_test))
- SQL: GROUP BY Landing_Outcome ORDER BY COUNT(*) DESC; SUM / AVG / MIN / MAX queries

Repository with all notebooks and Python files:

- [Paste your GitHub URL here]

Thank you!

